



The ICPP Protocol

A Platform for P2P Privacy-Enabled ICP Transfers

Brief Introduction and Usage Guide

v2.1.00

January 2026



Welcome to ICPP

This document explains the operation of ICPP, a new privacy protocol on the Internet Computer, in a non-technical and intuitive way. ICPP denotes privacy-enabled ICP.

Users of the Internet Computer (ICP) infrastructure and ICP token holders have in ICPP a means to execute P2P transactions using high-end cryptography (e.g., encrypted transport tunnels and capsules) alongside other unique features. ICPP also incorporates ephemerality. **Competing protocols are only just beginning to explore this approach for privacy, but ICPP makes the technology available today to the ICP community.**

What does ephemerality mean? In our context, it refers to using short-lived infrastructure layers or code components (in the case of the ICP, canisters) to process P2P transactions.

To understand this, let's use an analogy. Alice needs to cross a river to meet Bob, who is waiting on the opposite shore. She finds a bridge and crosses it, but as soon as she reaches the other side, the bridge vanishes. She cannot retrace her steps and the physical evidence of her crossing has disappeared. No bridge remains, nor is there any visible trace suggesting how Alice could have plausibly reached the other side.



Is ICPP a new Token?

That's the next, and quite relevant, question many potential users would be inclined to ask, for which there is a simple and direct answer.

Transactions using ICPP involve vanilla ICP tokens.

It means there is no need for an ad-hoc token, or for wrapped or bridged variants. Just plain ICP. The platform allows anyone with an ICPP account to make shielded P2P ICP transfers to other ICPP account holders. **The beneficiary will see ICP tokens, identical to every other ICP token, credited in their account and ready for use.**

The upside?

Transactions on ICPP leave no meaningful traces. At most, only encrypted breadcrumbs.

How, you may ask?



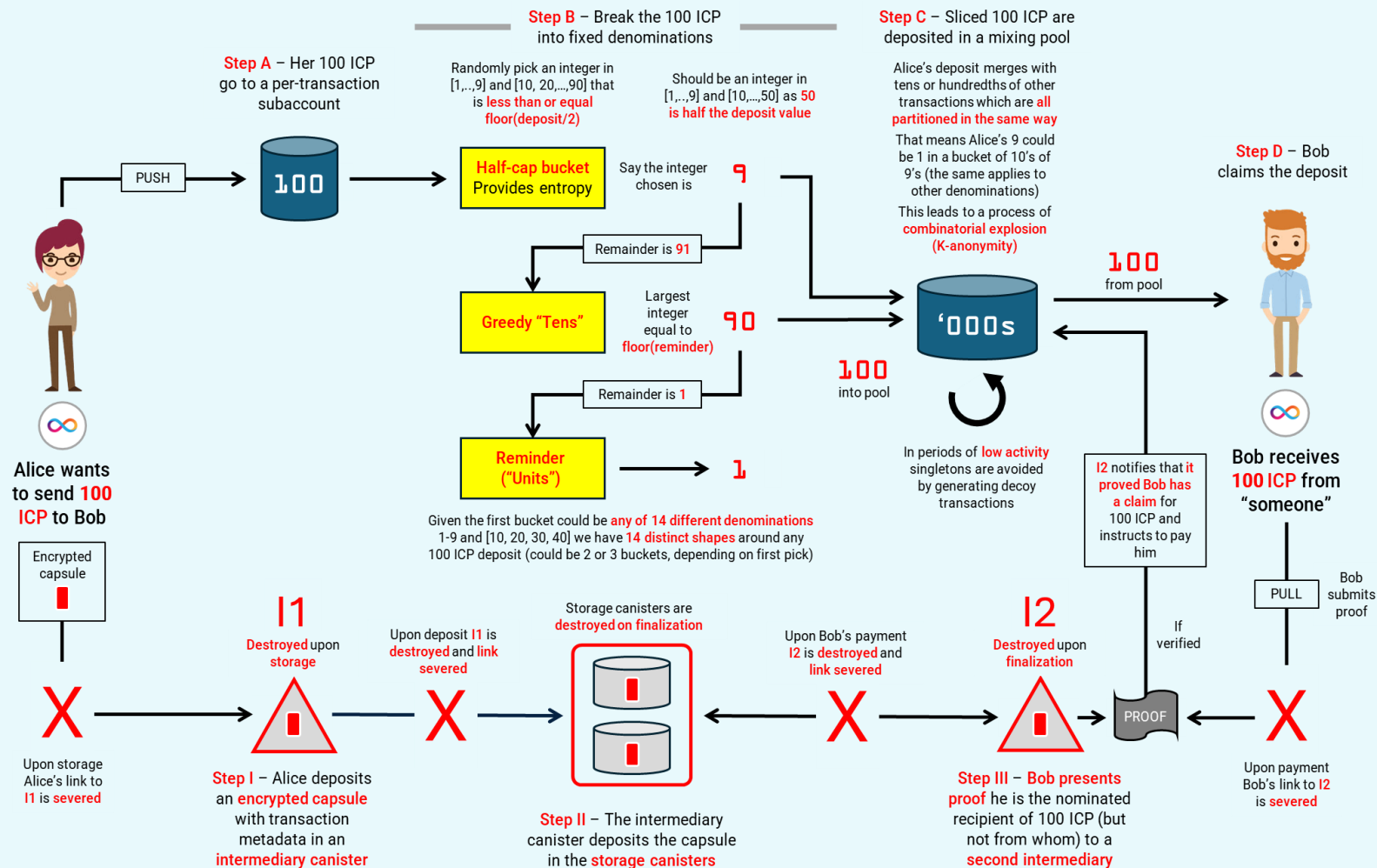
The mechanics, in plain language

Let's say Alice wants to transfer some ICP to Bob.

- The canisters processing P2P transactions, which are spawned on a per-transaction basis, are compartmentalized. **No single canister ever holds both Alice and Bob's identities simultaneously.**
- Besides a Router, a canister Factory, and a Noticeboard canister, each holding no information fully connecting Alice with Bob, **all other canisters intervening in an ICPP-mediated transaction are ephemeral.** Once the transaction completes, they are destroyed. **The partial knowledge each held is gone.** Even if it were possible to reconstruct any canister's state, none ever contained the full picture simultaneously. **The link between Alice and Bob never persisted in any single place.**
- **Transactions are pooled.** Alice's transfer to Bob is split into pieces and mixed with everyone else's transfers. Bob withdraws a single amount. An observer sees fragments going in and whole amounts coming out. **Matching them requires guessing by brute force which fragments belong together.**
- **Random time decorrelation** also contributes to hardening privacy, simply because Bob has to act (within a time window) upon knowledge of the transfer.



But how does it work, actually?



ICPP utilizes an infrastructure of permanent and ephemeral canisters to **decouple deposits from claims**. Through advanced cryptography and algorithmics, transaction data is processed so **the Protocol never associates a sender with their recipient**. In this way, ICPP preserves absolute privacy and on-chain unlinkability.

ICPP uses ICP's public ledger but no information is leaked linking Alice to Bob

¹ See <https://arxiv.org/abs/2512.23535>



Why ephemerality?

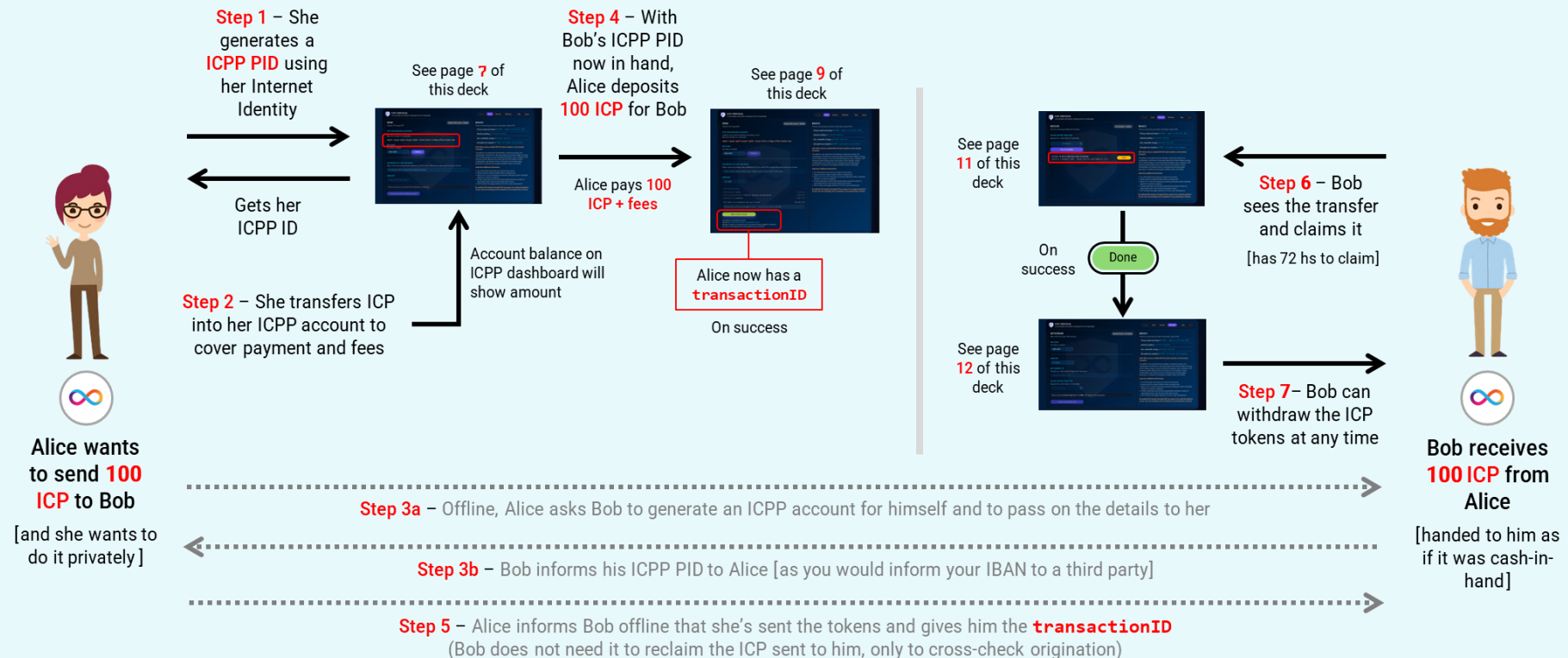
Ephemerality (infrastructure components that are destroyed) **bounds the adversary's computational window**. But it's also a **cryptographic property**, not just operational hygiene.

- Traditional crypto assumes an adversary captures ciphertext, has unlimited time to attack, and security must hold indefinitely or until key expiry. **Because encrypted transactions are stored forever on-chain.**
- In comparison, an adversary must attack IPCC during a **very limited canister lifetime**, leaving no persistent cyphertext to attack offline, and no keys to exfiltrate later.
- **This is forward secrecy applied at the architectural level.** Even if adversary compromises the system **after transaction completion**, they can't decrypt past transactions because the ephemeral state is cryptographically gone.
- As no single component in IPCC holds complete information, **an attacker must compromise multiple canisters simultaneously**. Doing so **under time pressure** (which compounds across components).
- To recap: IPCC is a **time-bounded, compartmentalized key agreement protocol** that leverages the IC unique properties and cryptography to create privacy guarantees that traditional blockchain privacy protocols cannot achieve.



Anatomy of a Transaction using ICPP

No need for command-based instructions. A visual interface makes it simple and stress-free. Online at <https://icpp.tech>





The ICPP Platform UX

We designed the ICPP interface to be as **easy, intuitive, and secure** as possible.



For access, go to <https://icpp.tech> hosted on the Internet Computer.

After **choosing a PIN** you will be directed to **DFINITY Internet Identity** gateway.

Proceed to link your IC account to ICPP. If successful, you will be redirected to the dashboard.

Credentials are device and browser specific. The combo generates a **locally-stored encrypted NK** that gates access to the platform.

Important: The ICPP dashboard only works in a **desktop or laptop devices**.



The ICPP Platform UX

Sending ICP using ICPP is quite simple

It involves providing some **basic information** common to any assets transfer

- 1** This is your **unique ICPP account ID**
[Transfer to this ICRC-1 account enough ICP tokens to provision the transactions you intend to make]
- 2** **Balance** of your ICPP account
[ICP tokens available to use]
- 3** **Destination** for the ICP transfer
[ICPP account of the intended recipient]
- 4** The **amount of ICP tokens** to send
[up to 2,000 ICP per transaction]

ICPP PROTOCOL
A PLATFORM FOR PRIVACY-ENABLED P2P ICP TRANSFERS

Access **Send** Receive Withdraw Help Logout

SEND
Transfer ICP using ICPP
Between ICPP accounts – Shielded

ICPP ORIGINATOR ACCOUNT
Linked to your Internet Identity and unique to you
Keep this number in a safe place
vmylt-rjynp-vpu7o-mcyy5-ajk5r-lisnv-b7u1v-sfnqq-of5jo-bej4u-4ae

BALANCE
ICP tokens available
1344.8442 Refresh

TRANSFER TO ICPP ACCOUNT
Please note that sending funds shielded by ICPP to a non-ICPP account will result in the loss of funds
53-character ICPP account of the intended recipient

AMOUNT
Maximum of 2,000 tokens
Enter an amount to check the fee breakdown
Review the transfer details and send

BASICS
Information about ICPP
SgpFE (v2.0.03 Dec 2025)
User credentials storage → Browser-local NK
Decryption by recipient → RDMPF (in-browser and isolated)
ICPP offers privacy-enabled P2P ICP token transfers on the Internet Computer.
The platform uses ephemeral intermediaries, transaction pooling, and cryptographic mechanisms to decouple deposit and retrieval stages, ensuring secure sealed storage in transit and attested teardown upon completion. ICPP preserves sender identity privacy with respect to the recipient, content confidentiality, and forward secrecy of transport keys through staged destruction. It achieves this alongside verifiable liveness and finality.
Important additional information
• Your PIN and NK never leave (nor leak from) this device.
• Always verify the recipient address before sending funds.

SEND tab

1

2

3

4

The recipient's Principal ID travels encrypted inside the inner payload of a capsule containing cryptographic material. On its way to destination, any checks (computations) done against the capsule's content do not require revealing the payload. Only the intended recipient can eventually open the capsule by deriving the correct key and prove its claim on the tokens.



The ICPP Platform UX

Complete transparency on transaction costs

ICPP ORIGINATOR ACCOUNT

Linked to your Internet Identity and unique to you
Keep this number in a safe place

vmylt-rjynp-vpu7o-mcyy5-ajk5r-lisnv-b7ulv-sfnqq-of5jo-bej4u-4ae

BALANCE

ICP tokens available

1344.8442

Refresh

TRANSFER TO ICPP ACCOUNT

Please note that sending funds to an incorrect address will result in the loss of funds

t25ys-ejeem-emo62-yhxcg-...

AMOUNT

350.0000

FEE BREAKDOWN

ICPP service fee → 0.5%	1.75 ICP
Cost of cycles provisioning → Canister spawning and operations	0.084 ICP
Ledger fees → 7 transfers	0.0007 ICP
ICP tokens to be deducted from your account	351.8347 ICP

Recipient gets credited exactly 350 ICP tokens | Transaction fees represent 0.524%

Review the transfer details and send

Privacy model and release → RDMPF + SgpFE (v2.0.03 Dec 2025)

Identity handling → Internet Identity

User credentials storage → Browser-local NK

Decryption by recipient → RDMPF (in-browser and isolated)

ICPP offers privacy-enabled P2P ICP token transfers on the Internet Computer.

The platform uses ephemeral intermediaries, transaction pooling, and cryptographic mechanisms to decouple deposit and retrieval stages, ensuring secure sealed storage in transit and attested teardown upon completion. ICPP preserves sender identity privacy with respect to the recipient, content confidentiality, and forward secrecy of transport keys through staged destruction. It achieves this alongside verifiable liveness and finality.

Important additional information

- Your PIN and NK never leave (nor leak from) this device.
- Always verify the recipient address before sending funds.
- Allow 10 to 15 min for transaction processing, although duration ultimately depends on network congestion.
- Canister spawning costs are fixed, progressively lowering its impact (in percentage terms) as the amounts being sent increase.
- ICPP has been thoroughly tested but it's still in active development.

By making P2P transfers through ICPP you agree to be using the platform at your own risk including, but not limited to, the potential loss of funds.

Get a real-time
cost estimate

ICPP adopts a Wise-style pricing structure. **The sender pays platform fees and operational costs** (canister spawning and ledger fees).

The **beneficiary** receives the **net amount of ICP tokens** being sent.

Platform fees are

0.5%

For the time being, ICPP sets a **cap** of **2,000 ICP tokens per transaction** to prevent abuse, whale crowding-out, and money laundering.



The ICPP Platform UX

Outbound transactions demand about 10 to 15 minutes for confirmation (sometimes less)

Transactions on ICPP are final

Once a transaction is executed, the only way to claw back the tokens is to ask the recipient for their reimbursement.

[we are currently exploring mechanisms for reverting transactions without compromising privacy]

The screenshot displays the ICPP Protocol web interface. At the top, it says 'ICPP PROTOCOL' and 'A PLATFORM FOR PRIVACY-ENABLED P2P ICP TRANSFERS'. The main section is titled 'SEND' and shows a balance of 1268.0163 ICP tokens. A red box highlights the 'Transfer ID' and the 'Recipient will get an encrypted hint of transaction being available for collection' message. A red box also highlights the 'Make another transfer' button. The right sidebar contains 'BASICS' information, including 'Privacy model and release', 'Identity handling', 'User credentials storage', and 'Decryption by recipient'. A red box highlights the 'Important additional information' section, which lists several bullet points and a disclaimer.

SEND
Transfer ICP using ICPP

ICPP ORIGINATOR ACCOUNT
Linked to your Internet Identity and unique to you
Keep this number in a safe place
vmylt-rjynp-vpu7o-mcyy5-ajk5r-lisnv-b7ulv-sfnqq-of5jo-bej4u-4ae

BALANCE
ICP tokens available
1268.0163

UPON CONFIRMATION YOU WILL BE NOTIFIED AND ISSUED A TRANSFER ID [keep this information safely for future reference]

Make another transfer

Transaction successfully processed
Transfer ID bc4d642db8c42853c1e6de005f6fc2e5
Recipient will get an encrypted hint of transaction being available for collection
This hint is both ephemeral and unlinkable to your account ID

BASICS
Protocol versioning and other information about ICPP

Privacy model and release → RDMPF + SgPFE (v2.0.03 Dec 2025)

Identity handling → Internet Identity

User credentials storage → Browser-local NK

Decryption by recipient → RDMPF (in-browser and isolated)

ICPP offers privacy-enabled P2P ICP token transfers on the Internet Computer.

The platform uses ephemeral intermediaries, transaction pooling, and cryptographic mechanisms to decouple deposit and retrieval stages, ensuring secure sealed storage in transit and attested teardown upon completion. ICPP preserves sender identity privacy with respect to the recipient, content confidentiality, and forward secrecy of transport keys through staged destruction. It achieves this alongside verifiable liveness and finality.

Important additional information

- Your PIN and NK never leave (nor leak from) this device.
- Always verify the recipient address before sending funds.
- Allow 10 to 15 min for transaction processing, although duration ultimately depends on network congestion.
- Canister spawning costs are fixed, progressively lowering its impact (in percentage terms) as the amounts being sent increase.
- ICPP has been thoroughly tested but it's still in active development.

By making P2P transfers through ICPP you agree to be using the platform at your own risk including, but not limited to, the potential loss of funds.



The ICPP Platform UX

Receiving ICP tokens using ICPP is equally straightforward

The screenshot shows the ICPP Protocol dashboard. At the top, there's a navigation bar with 'Access', 'Send', 'Receive', 'Withdraw', 'Help', and 'Logout'. The 'Receive' tab is highlighted with a red box and labeled 'RECEIVE tab'. Below the navigation bar, the 'RECEIVE' section is titled 'RECEIVE' and 'A PLATFORM FOR PRIVACY-ENABLED P2P ICP TRANSFERS'. It includes a 'Local decryption - Shielded' button. The main area prompts the user to 'PLEASE ENTER YOUR PIN' and 'Required for confirmation of ownership'. There's a text input field with '6 or more digits' and a 'Scan for transfers' button, both highlighted with a red box. A red callout box points to this area, stating: 'As a security precaution you are asked to re-enter your PIN. This allows you to scan the platform's Noticeboard for encrypted hints'. To the right, the 'BASICS' section provides information about the protocol version, privacy model, identity handling, user credentials storage, and decryption methods. It also includes a disclaimer about the platform's use and a list of important additional information.

Simply **provide your PIN** for the dashboard to **scan the ICPP Noticeboard** containing hints.

Users **attempt decryptions** on those announcements (using an efficient algorithm). Success identifies which transaction(s) have you as beneficiary.

The reclaim process is very easy.

Similarly to when you send ICP using ICPP, when claiming funds the process can take **10 to 15 minutes** (depending on network congestion).



The ICPP Platform UX

Now just wait for any ICP token sent to you to land on your IC account

ICPP PROTOCOL
A PLATFORM FOR PRIVACY-ENABLED P2P ICP TRANSFERS

Access Send **Receive** Withdraw Help Logout

RECEIVE
Scan for incoming shielded ICP transfers
Local decryption - Shielded

PLEASE ENTER YOUR PIN
Required for confirmation of ownership

.....

Scan for transfers

Transfer ID 00cf1c260a919eac72051c1e910ef98d
Stored in 2 ephemeral nodes - Please note the claim window is 72 hs

Claim

Transfer ID 00cf1c260a919eac72051c1e910ef98d
Claiming funds - Usually takes 10 to 15 min for confirmation

Processing

Transfer ID 00cf1c260a919eac72051c1e910ef98d
Success - Check the balance of your ICP accounts

Done

BASICS
Protocol versioning and other information about ICPP

Privacy model and release → RDMPP + SgPFE (v2.0.03 Dec 2025)

Identity handling → Internet Identity

User credentials storage → Browser-local HK

Decryption by recipient → RDMPP (in-browser and isolated)

ICPP offers privacy-enabled P2P ICP token transfers on the Internet Computer. The platform uses a staged approach to ensure security and privacy. During the claim stage, the recipient's identity is preserved. Once the claim is complete, the funds are transferred to the recipient's account. ICPP preserves sender identity privacy with respect to the recipient, content confidentiality, and forward secrecy of transfers. Funds are destroyed after a finality period.

- You must always verify the recipient address before sending funds.
- Allow 10 to 15 min for transaction processing, although duration ultimately depends on network conditions.
- Canis is a privacy-preserving protocol that ensures the confidentiality of transactions.
- ICPP is a privacy-preserving protocol that ensures the confidentiality of transactions.

By making P2P transfers through ICPP you agree to be using the platform at your own risk including, but not limited to, the potential loss of funds.

If you have a transfer pending you can CLAIM it
Ask the sender to provide you the Transaction ID [using an offline side channel] for you to cross-check

Press CLAIM

Wait to PROCESS

SUCCESS

Important: On very rare instances a transaction could get stuck. These include (but are not limited to) **loss of connection, II identity and delegation staleness, or IC network hiccups**. The platform has several in-built failure prevention features. As first port of call, wait until the **TRY AGAIN** option becomes available. **If that doesn't work, restart your browser, login and rescan**. Operationally, if you left the IPCC dashboard sitting idle for **more than 30 min** we recommend to log out and then back in to ensure delegation is refreshed before attempting any action.



The ICPP Platform UX

Withdraw ICP tokens on ICPP to your account of choice for immediate use

The screenshot shows the ICPP Protocol withdrawal interface. At the top, there's a navigation bar with 'Access', 'Send', 'Receive', 'Withdraw' (highlighted with a red box and labeled 'WITHDRAW tab'), 'Help', and 'Logout'. The main section is titled 'WITHDRAW' and 'A PLATFORM FOR PRIVACY-ENABLED P2P ICP TRANSFERS'. It includes a 'BALANCE' section showing 'ICP tokens available' as '1268.0163' (step 1). Below that is an 'AMOUNT' section with a dropdown menu set to 'ICP tokens' (step 2). The 'WITHDRAW TO' section asks for a '53-character ICRC-1 PID' (step 3). The 'PLEASE ENTER YOUR PIN' section requires a '6 or more digits' PIN (step 4). At the bottom, there's a note about the standard ledger fee of 0.0001 ICP and a 'Confirm and withdraw ICP' button. A sidebar on the right contains 'BASICS' information about the protocol, including versioning, privacy models, identity handling, and user credentials storage.

- 1 **Balance** of your ICPP account
[ICP tokens available for extraction]
- 2 Decide on the **amount of ICP tokens** to move out
[Transfer to any account supporting the ICRC-1 standard]
- 3 **Recipient** of the ICP tokens from ICPP
[Account where the tokens will be sent to]
- 4 **Enter your PIN** and withdraw the tokens
[with no caps or other restrictions]



**Making ICP transactions
private on the Internet
Computer**

<https://icpp.tech>

icpp.tech@proton.me